



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SANJAY FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

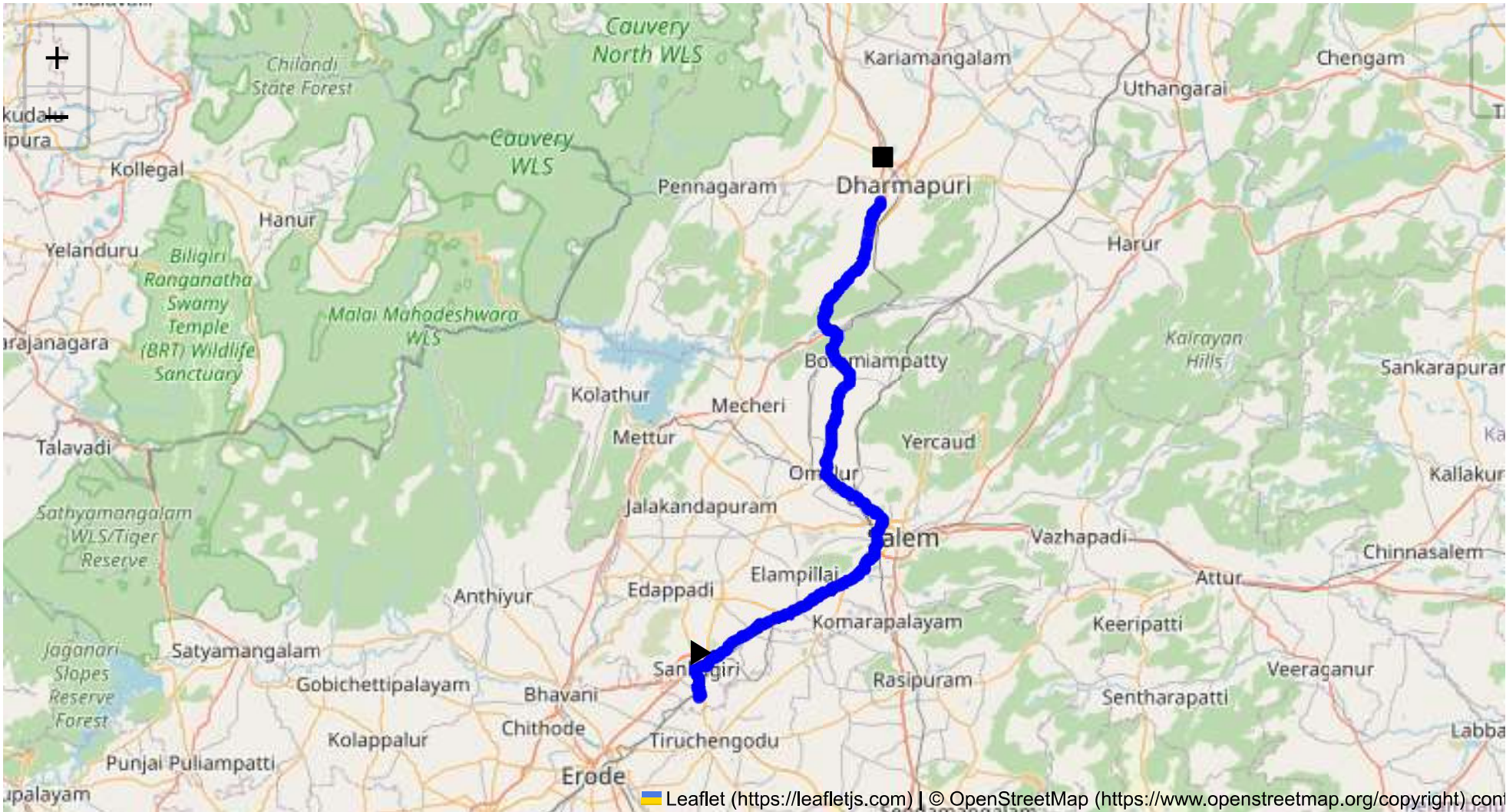
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 101.83 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.5 hours
Start: (11.4381, 77.8734)
End: (12.110828, 78.126273)



Welcome to the Journey Risk Management Study

Route Safety Analysis: CVQF+23W, Sangagiri, Tamil Nadu to 27/1 Thadangam, Tamil Nadu

1. Overview of the Route Map

The route spans approximately 101.83 kilometers, typically taking about 1.97 hours for heavy vehicles. The journey starts from Sangagiri and proceeds through state highways and possibly stretches of NH 44. The route offers a mix of rural and semi-urban driving conditions with diverse infrastructure quality.

2. Typical Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate with seasonal variations:

- **Summer (March - May):** High temperatures can lead to vehicle overheating and asphalt softening.
- **Monsoon (June - September):** Heavy rains may cause flooding and slippery roads, increasing the risk of hydroplaning.
- **Winter (October - February):** Mild temperatures, but occasional fog in early mornings that might affect visibility.

3. Analysis of Traffic Patterns

- **Peak Hours:** Generally, 8 AM–10 AM and 5 PM–8 PM are peak traffic times due to local commutes.
- **Congestion-Prone Areas:** Town centers and market areas in both Sangagiri and Thadangam. NH routes might experience delays especially due to checkpoints and toll booths.

4. Assessment of Road Quality and Infrastructure

- **State Highways:** Variable road quality; sections may have potholes or uneven surfaces.
- **National Highways:** Generally better quality but watch for construction zones.
- **Bridges/Rail Crossings:** Ensure to be cautious near bridges and railway crossings as they may have delays or structural challenges.

5. Alternate Routes for Emergencies

- Consider using alternative rural roads parallel to the national highway for bypassing long congestion sections if the primary route is blocked.
- Route via SH 20 or SH 81 can be considered for diversion, depending on the specific problem faced.

6. Local Regulations

- Tamil Nadu mandates permits for hazardous materials transport.
- Speed limits for trucks are strictly enforced, especially on NHs.
- Regulations on vehicle weight and dimensions should be adhered to strictly.

7. Historical Incidents

- There have been recorded incidents involving overloaded trucks and hazardous spills mainly near busy junctions on the National Highway.
- Precautions should be taken seriously where past incidents have occurred.

8. Environmental Considerations

- Several small water bodies and agricultural lands may be adjacent to roads; any spills or incidents can have high environmental impacts.
- Observe care near wildlife crossings and reserve areas en route.

9. Communication Coverage

- Generally good along highways and urban areas.
- Potential dead zones in rural stretches; therefore, ensure backup communication systems are available.
- Use updated navigation systems with offline capabilities.

10. Emergency Response Times

- **Urban Areas:** Fast emergency response, usually within 15–30 minutes.
- **Rural Stretches:** Delays in response can be expected, likely 45 minutes to 1 hour.
- Insufficient medical facilities along some rural parts of the route; plan appropriately.

11. Overall Summary of Risk Assessment

This route involves moderate risk levels primarily due to:

- Traffic congestion potential and poor road conditions on some segments.
- Monsoon-related hazards affect road reliability.
- Possible delays in emergency response and communications challenges.
- Sound planning and adherence to regulations are crucial for mitigating these risks.

Overall, drivers should ensure complete vehicle checks, weather forecasts monitoring, and communication devices are operational before embarking on this route with hazardous materials. Being aware of historical incident spots and potential environmental impacts due to spills is critical for safe transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	Medium	11.45000, 77.87213	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.45309, 77.87325	30 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	Medium	11.46765, 77.87037	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.64180, 78.11808	30 KM/Hr
12	Turn	Medium	11.64325, 78.11993	30 KM/Hr
13	Turn	Medium	11.69519, 78.10775	30 KM/Hr
14	Turn	Medium	11.75835, 78.05016	30 KM/Hr
15	Turn	Medium	11.95818, 78.04978	30 KM/Hr
16	Turn	Medium	11.95876, 78.04990	30 KM/Hr
17	Turn	Medium	11.96035, 78.04946	30 KM/Hr
18	Turn	Medium	11.96304, 78.05136	30 KM/Hr
19	Turn	Medium	11.96399, 78.05135	30 KM/Hr
20	Turn	Medium	11.96458, 78.05084	30 KM/Hr
21	Turn	Medium	11.96526, 78.04950	30 KM/Hr
22	Turn	Medium	11.96580, 78.04923	30 KM/Hr
23	Turn	Medium	11.96675, 78.04939	30 KM/Hr
24	Turn	Medium	11.96791, 78.05070	30 KM/Hr
25	Turn	Medium	12.09474, 78.11666	30 KM/Hr
26	Turn	Medium	12.09489, 78.11665	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
32	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
33	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
35	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
36	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
37	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
38	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
39	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
40	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
24	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
30	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
31	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
34	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
41	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49224, 77.89601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64180, 78.11808



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64325, 78.11993



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69519, 78.10775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75835, 78.05016



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95818, 78.04978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95876, 78.04990



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96035, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96458, 78.05084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96526, 78.04950



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96675, 78.04939



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96791, 78.05070



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.09474, 78.11666



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.09489, 78.11665